14.01 – Principles of Microeconomics

Midterm Exam

Fall 2023

1 True or False (10 Points)

True or False? Justify your answer. Correct answers without the proper justification will not receive credit.

1. *(5 Points)* Suppose a consumer's preferences over goods x and y is $U(x,y) = min\{x,y\}$. Then, if $p_x > p_y$ he does not consume any of y and spends all of his income on good x.

   False. for the consumers utility function is: L, thus they are perfect complements for them. so he buys equal on both

2. *(5 Points)* Suppose the market for shoes is composed by identical firms with production function given by $q = L$, where L stands for labor. Suppose in the short run there are J firms. Then, the number of firms in the long run is also equal to J.

   True, $q=l$ is constant returns to scale so profit=0 thus noone enters the market

2 Consumer Theory (35 Points)

Pedro has preferences over two goods: food x and attending the Boston Celtics' games y. His preferences are given by

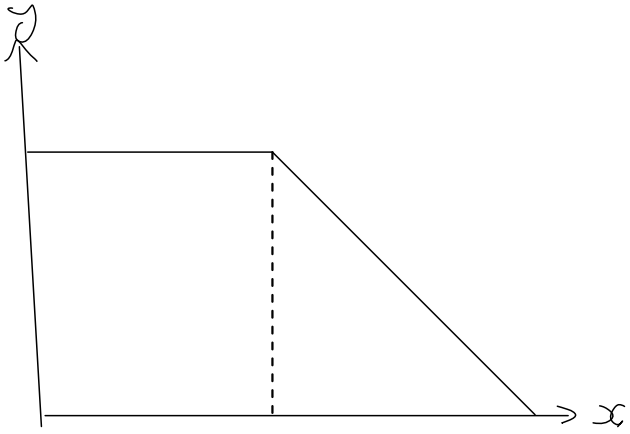
$$U(c,l) = x^{\frac{3}{4}} y^{\frac{1}{4}}$$

Food has a price p_x and a ticket to attend a Boston Celtics' game costs p_y dollars. Pedro's income is equal to m dollars. Pedro is also eligible for f dollars in Food Stamps. Food stamps is additional income that can only be spent in food. Throughout this exercise assume that $f < 3m$

1. *(3 Points)* Write down and graph Pedro's budget constraint.

$p_x x + p_y y = m + f$ if $p_x x$ is more than f

$p_y y = m$ if $p_x x$ is less than f



2. *(5 Points)* Calculate Pedro's demand for food and tickets, as a function of the prices p_x and p_y, his income m and the value of food stamps f.

$MU_x = 3/4 \cdot x^{-1/4} \cdot y^{(1/4)}$ $MU_y = 1/4\, x^{3/4} \cdot y^{-3/4}$

$MRS = 3y/x = \frac{P_x}{P_y}$ $P_y \cdot y = P_x x/3$

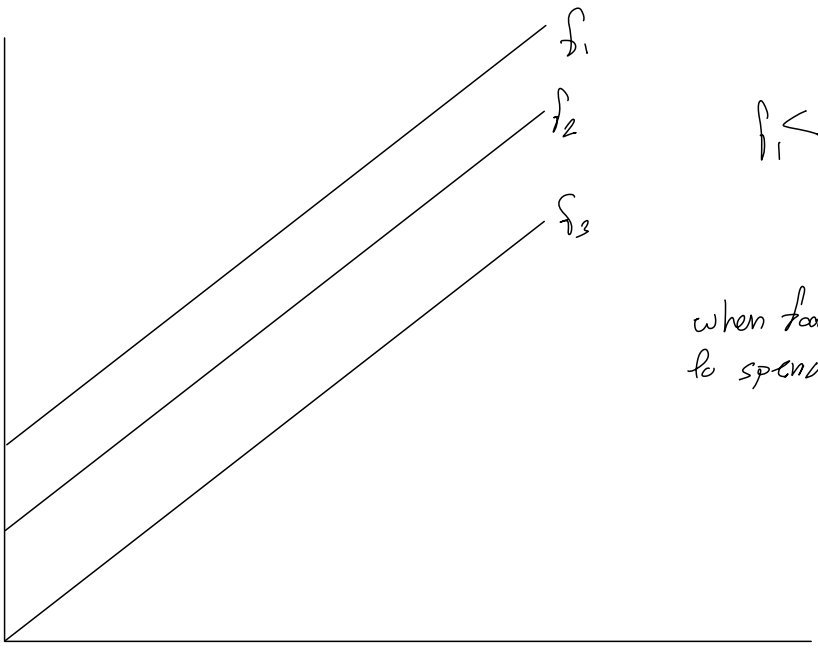
$P_x \cdot x + P_x \cdot x/3 = m+f$ $\cdot (4/3) \cdot P_x x = m+f$

$x = 3(m+f)/(4P_x) \longrightarrow \frac{3m+3f}{3} > f$ because $3m > f$

$y = (m+f)/(4P_y)$

3. *(5 Points)* Graph Pedro's Engel curve for food for different values of food stamps f and assuming $p_x = 1$. Provide an intuition on how the Engel curve depends on the value of food stamps.

for the engel curve $m = 1/3\, x - f$ $\therefore$ with f increasing it goes down



$f_1 < f_2 < f_3$

when food stamps can get more x he needs less m to spend on them thus the engel curve goes down

Suppose Pedro's income is $m = 1000$ dollars and he receives food stamps for $f = 200$ dollars. Prices are initially $p_x = 1$ and $p_y = 1$. Suddenly, the price of food increases to $p'_x = 2$.

4. *(10 Points)* Calculate Pedro's demand for food and tickets before and after the change in prices. Decompose the effect of a change in prices into substitution and income effect. Graph Pedro's consumption choices, as well as the decomposition into substitution and income effect.

$p_x = 1$

$x = 900$

$y = 300$

$p'_x = 2$

$x = 450$

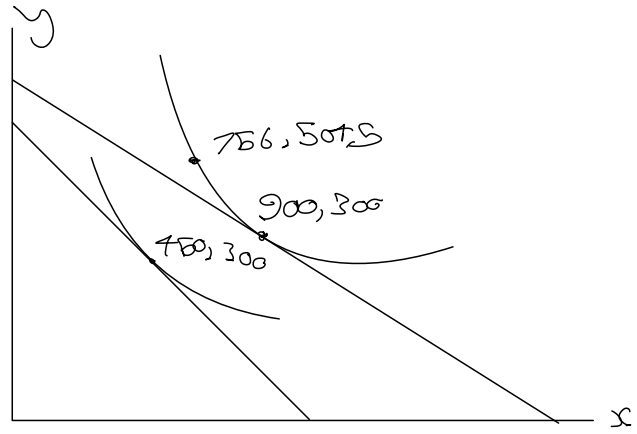
$y = 300$

$u = 900^{3/4} \cdot 300^{1/4} \approx 683.85$

$x^{3/4} \cdot \left(\frac{2x}{3}\right)^{1/4} = 683.85 \quad x = 756.8, \; y = 504.5$

$900 - 756.8 = -143.2$) substitution effect

$756 - 450 = -306.8$) income effect



5. *(5 Points)* Calculate the increase in food stamps necessary for Pedro's utility to be unchanged after the change in prices.

$x = 3(1000 + f)/8 = 900$

$3000 + 3f = 7200$

$3f = 4200$

$f' = 1400$ f has to go to 1400

6. *(2 Points)* Suppose instead of food stamps Pedro receives a stimulus check from the Federal Government. Let s denote the size of the stimulus check. What are the new demands for food and tickets as a function of prices p_x and p_y, his income m and the stimulus check s

$p_x x + p_y y = m + s$

$x = \frac{3m + 3s}{4p_x}$ $y = \frac{m+s}{4p_y}$

7. *(5 Points)* What should be the size of the stimulus check s for Pedro to be indifferent between receiving food stamps or the stimulus check?

$s = f$, pedro spends more on food than the stimulus checks anyways so it would all go into food again.

3 Short-Run Producer Theory (15 Points)

You own an ice cream factory and employ both capital K and labor L to produce ice cream. Capital is fixed at $0 < \bar{K} < 1$. Your production function is

$$F(L, \bar{K}) = \log(1 + L) + \log(1 + K)$$

where log is the natural logarithm i.e., $\log x = y \iff x = e^y$. Recall that $\log(1) = 0$.

Suppose that you have no capital in the short run (so $\bar{K} = 0$).

1. *(3 Points)* Find the short-run cost function, as a function of w, r, $\bar{K}$ and q.

$\log(1+L) = q - \log(1+k)$

$1+L = e^{(q - \log(1+k))} = e^{\log(q/1+k)} = e^{q}/(1+k) - 1$

$C(q) = w(e^{q/(1+k)-1}) + r \cdot k$

2. *(5 Points)* Find the short-run marginal cost and average cost. If the firm chooses to produce, how much will it produce as a function of the price?

$MC = (q-1) \cdot w \cdot e^{q-2}$

$AC = \frac{w(e^{q-1})}{q}$ it will produce when $MC > AC$

3. *(7 Points)* Now suppose that $\bar{K}$ can be chosen to minimize cost for given prices and quantity produced. Suppose that $w = r = 1$. Solve for the firm's long run cost function and long-run supply curve.

$$MRTS = \frac{-w}{r}$$

$$1 + lc = w/r\,(1 + L)$$

$$\log(1+L) + \log\left(\frac{w}{r}(1+L)\right) = q$$

4 Long-Run Production Theory (30 Points)

Now suppose that instead there is a different ice cream factory that the following long-run cost function

$$C(q) = \begin{cases} q^2 + 9 & q > 0 \\ 0 & q = 0 \end{cases}$$

Suppose that the market for ice cream is competitive, and firms take prices as given.

1. *(3 Points)* Calculate the firm's long run supply curve. Does the firm choose to produce a positive quantity at any price? yes

   $MC = 2q$

   $p = 2q$

Suppose there are 4 identical firms producing ice-cream in the town. These firms hold ice-cream production licenses so that the number of firms is at most 4. The market demand for ice-cream is

$$Q_D = 30 - p$$

2. *(2 Points)* Write down the aggregate supply of ice cream.

   $Q_S = 4p \quad p \geq 6$

   $Q_S = 0 \quad p < 6$

3. *(5 Points)* Find the equilibrium price and quantity, as well as the profit made by each firm.

$p^* = 15$, $Q = 120$

$\text{Profit} = \frac{(15 \cdot 120)}{4} - 129 = 321$

Suppose that a firm has discovered a new recipe for ice cream without lactose. Since most of the town is lactose intolerant the government decides to shut down the production of ice cream with lactose. Since this ice cream is healthier, people demand it more. In particular, the new demand is

$$Q_D = (30 + A) - p$$

where $A > 0$ is an exogenous constant that reflects the higher popularity of lactose-free ice cream. Only one firm has the patent for this new recipe, and has the same production technology stated above, namely $C(q) = q^2 + 9$ if $q > 0$ and 0 if $q = 0$.

4. *(5 Points)* Solve for the monopolist's optimal choice of quantity and price as a function of A.

$C(q) = q^2 + 9$

$Q_D = (30 + A) - p$

$R = q \cdot p = mc$

$mc = 2q$

$R = q(30 + A - q)$

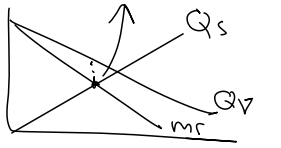
$2q = (q(30 + A - q))\frac{dq}{dy} \therefore q^* = \frac{30 + A}{4}$, $p^* = \frac{90 + 3A}{4}$

5. *(5 Points)* Compute the consumer and producer surplus in this market. Is it socially efficient? If not, compute the deadweight loss.

$$C_S = \int_0^{\frac{30+A}{4}} \left((30+A-q) - \left(\frac{90+3A}{4}\right) \right) dq = (30+A)^2/32$$

$$P_S = \int_0^{\frac{30+A}{4}} \left(\left(\frac{90+3A}{4}\right) - 2q \right) dq = \frac{(30+A)^2}{8}$$

Deadweigh



is it socially efficient? $\Rightarrow$ no

6. *(7 Points)* Are consumers better off consuming lactose-free ice cream even though there is a monopolist compared to a situation where there is competition but ice cream is not lactose-free? How does this depend on A? Provide an economic intuition. For full credit your answer must have an algebraic expression as well as an economic intuition. Comparing consumer surpluses without proper intuition will not result in full credit

$C_{SF} = 200$

$C_{SM} = \frac{(30+A)^2}{32}$

∴ if $A > 50$ then the surge in Demand offsets the monopolies $MR=MC$ Q and gives them more surplus.

Now consider a government who wishes to regulate the monopolist.

7. *(3 Points)* If the government can put a cap on the price of ice cream, what price cap should the government choose? Provide an economic intuition.

the government's goal would be to eliminate deadweight loss and make the market act like a free one

$\therefore P_{cap} = \frac{60+2A}{3}$

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